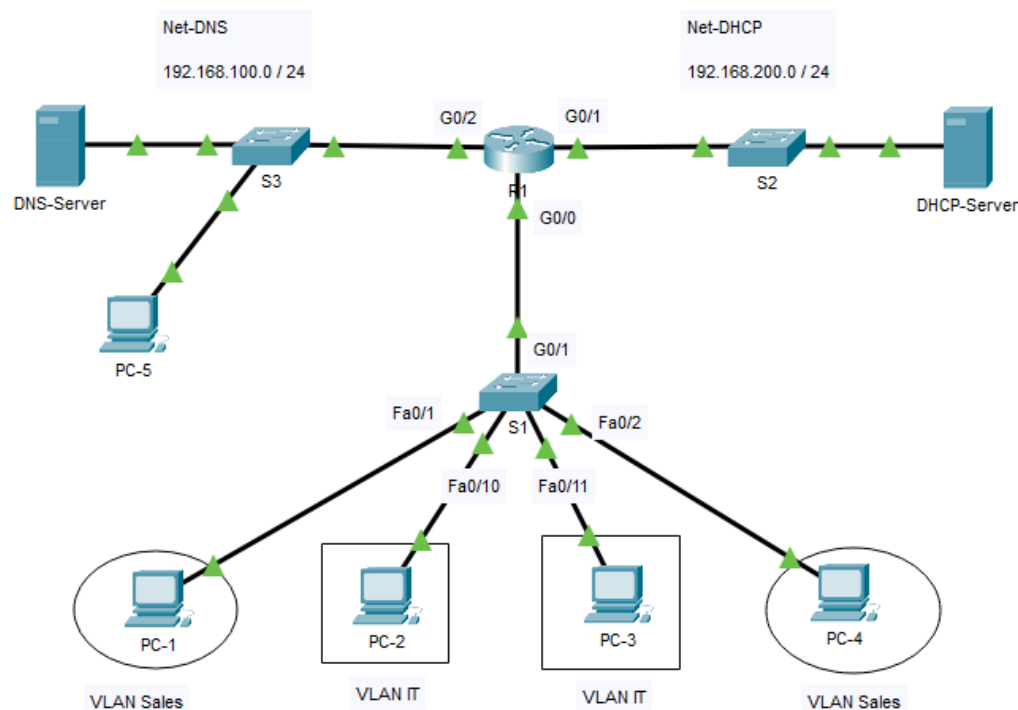


Question I:

	Answer
1) In a router-on-a-stick configuration, what type of port is used on the switch?	A. Access port B. Trunk port C. Routed port D. Loopback port
2) In an 802.1Q trunk link between two switches, what is the purpose of the native VLAN?	A. To carry both tagged and untagged traffic across the trunk link B. To carry untagged traffic across the trunk link C. To carry tagged VLAN traffic D. None of the above
3) What is required for devices in different VLANs to communicate with each other?	A. DHCP Server B. Access Ports C. Inter-VLAN Routing D. Port Security
4) What must be configured on a subinterface to enable communication with its respective VLAN?	A. Port Security B. IP helper-address C. Encapsulation dot1Q and an IP address D. NAT overload
5) Which Cisco router configuration command enables a subinterface for Inter-VLAN routing?	A. interface vlan X B. ip dhcp pool VLANX C. interface FastEthernet0/X.X D. vlan database
6) Which command would show if port security has been violated on a specific port?	A. show running-config B. show vlan security C. show port-security violation interface [interface-id] D. show port-security interface [interface-id]
7) Which of the following is NOT a valid port security violation mode?	A. Shutdown B. Restrict C. Protect D. Alert
8) Which of the following is TRUE about the protect mode in port security violation?	A. It drops packets from unknown MACs without generating logs B. It allows traffic but limits bandwidth C. It logs and drops packets from unknown MACs D. It disables the interface upon violation
9) Which of the following statements about Dynamic	A. DTP is used to automatically assign IP addresses to trunk ports B. DTP is required for static trunk configuration to work

Trunking Protocol (DTP) is TRUE?	C. DTP allows switches to negotiate trunk links automatically D. DTP encrypts VLAN traffic between switches
10) You configure a router-on-a-stick setup, but inter-VLAN communication still fails. Which of the following is NOT likely the cause?	A. The switchport connected to the router is in dynamic desirable mode B. Native vlan mismatch C. IP address is not assigned to the router D. Allowed vlan mismatch
11) Which global configuration command must be enabled on a Cisco multilayer switch to allow inter-VLAN routing functionality?	E. ip routing F. router ospf 1 G. switchport mode layer3 H. ip forward-protocol
12) What action does a Cisco switch port take by default when a Port Security violation is detected?	I. protect J. restrict K. shutdown L. disable

Question II:



Addresses Table:

Device	Interface	IP address	Default gateway
R1	G0/0	Subinterfacing	-
	G0/1	192.168.200.1	-
	G0/2	192.168.100.1	-
DHCP-Server	FastEthernet0	192.168.200.254	192.168.200.1
DNS-Server	FastEthernet0	192.168.100.166	192.168.100.1
PC-1	FastEthernet0	DHCP Assigned	192.168.10.1
PC-2	FastEthernet0	DHCP Assigned	192.168.40.1
PC-3	FastEthernet0	DHCP Assigned	192.168.40.1
PC-4	FastEthernet0	DHCP Assigned	192.168.10.1
PC-5	FastEthernet0	DHCP Assigned	192.168.100.1

VLAN	VLAN name	Ports on S1
10	Sales	Fa0/1-9
40	IT	Fa0/10-19
999	Native	

Networks	IP Addresses
Net-DHCP	192.168.200.0/24
Net-DNS	192.168.100.0/24
Net-Sales	192.168.10.0/24
Net-IT	192.168.40.0/24
Native	192.168.99.0/24

I- VLAN configuration

- 1) Give one advantage and one disadvantage of using Layer 3 switch for inter-vlan communication over using router-on-a-stick configuration. [6 pts.]

Advantage of using a Layer 3 switch: **Higher performance:** Layer 3 switches offer faster inter-VLAN routing because routing is done in hardware (ASIC), reducing latency compared to the software-based routing in a router-on-a-stick configuration.

Disadvantage of using a Layer 3 switch: **Higher cost:** Layer 3 switches are generally more expensive than standard Layer 2 switches and routers used in router-on-a-stick setups.

- 2) Why is there a need for a router to allow the devices in IT VLAN to communicate with the devices in Sales VLAN, even if they are both connected to the same switch? [4 pts.]

When a device wants to communicate with an IP address outside of its network (VLAN), it will send it to the MAC address of the default gateway - which is a router – instead of using the MAC address of the destination device.

- 3) Give the configurations that must be done on S1 to assign each port to its appropriate VLAN. [8 pts.]

S1(config)#Vlan 10

S1(config-vlan)#name Sales

S1(config)#Vlan 40

S1(config-vlan)#name IT

S1(config)#Vlan 999

S1(config-vlan)#name Native

S1(config)#int range fa0/1-9

S1(config-if)# switchport mode access

S1(config-if)# switchport access vlan 10

S1(config-if)#int range fa0/10-19

S1(config-if)# switchport mode access

S1(config-if)# switchport access vlan 40

- 4) Give the configuration that must be done on port G0/1 on S1 and on G0/0 on R1 to allow inter-VLAN communication. [12 pts.]

```
S1(config)#int G0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,40, 999
S1(config-if)#switchport trunk native vlan 999
```

```
R1(config)#int G0/0.10
R1(config-if)#enc dot1q 10
R1(config-if)#ip add 192.168.10.1 255.255.255.0
R1(config-if)#int G0/0.40
R1(config-if)#enc dot1q 40
R1(config-if)#ip add 192.168.40.1 255.255.255.0
R1(config-if)#int G0/0.999
R1(config-if)#enc dot1q 999 native
R1(config-if)#ip add 192.168.99.1 255.255.255.0
R1(config-if)#exit
R1(config)#int g0/0
R1(config-if)#no shut
```

- 5) Suppose that we have a new switch called S1-1 where the following ports are assigned in the same way as the ports on S1. We established a link from G0/1 on S1-1 to G0/2 on S1 to allow the members of both VLANs on S1 to communicate with the correspondent members on S1-1. Give the complete configuration that must be done on both sides of this link. Use the appropriate native VLAN. [6 pts.]

```
S1(config)#int G0/2
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,40, 999
S1(config-if)#switchport trunk native vlan 999
```

```
S1-1(config)#int G0/1
S1-1(config-if)#switchport mode trunk
S1-1(config-if)#switchport trunk allowed vlan 10,40, 999
S1-1(config-if)#switchport trunk native vlan 999
```

- 6) suppose that the members of Sales on S1 can communicate with the members of IT and Sales on S1, also the members of Sales on S1 can communicate with the members of sales on S1-1 but NOT the IT on S1-1. Where could the problem be? [4 pts.]

IT is not allowed on the trunk link between S1 and S1-1

- 1) Given the Data below, Enable the port security feature on the ports of S1 that are connected to PC-1, PC-2, and PC-3, ensure that each switchport can only be configured to maximum one address that will be learned dynamically and be saved permanently, any in case of violation a notification must be sent to syslog server. Finally, manually configure F0/2 to accept the MAC addresses of PC-3 and PC-4, apply the strictest action in case of violation [8 pts.]

PC	MAC Address	Connected To Port
PC-1	0001.97C7.E4AB	Fa0/1
PC-2	0001.6434.486D	Fa0/10
PC-3	0050.0F5E.3384	Fa0/11
PC-4	0001.9624.E703	Fa0/2

Switch(config)#int range fa0/1, fa0/10-11

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#exit

Switch(config)#int fa0/2
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 2
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address 0001.9624.E703
Switch(config-if)#switchport port-security mac-address 0050.0F5E.3384